

Cost-Effectiveness of Neoadjuvant Pembrolizumab + Chemotherapy in Early-Stage Triple-Negative Breast Cancer

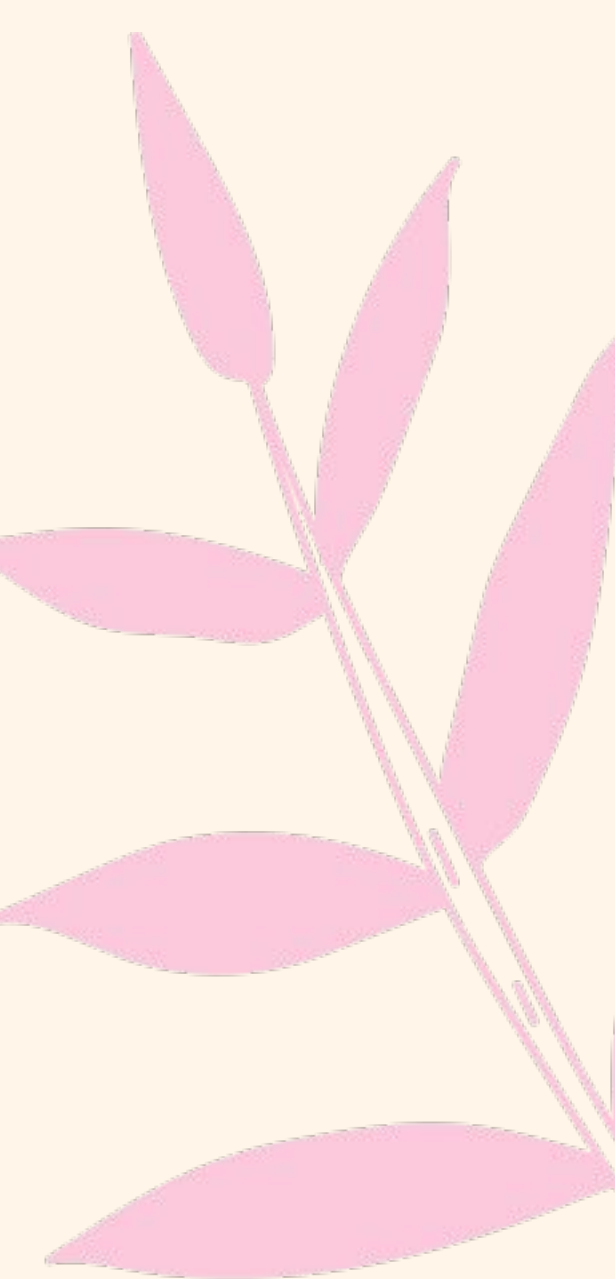
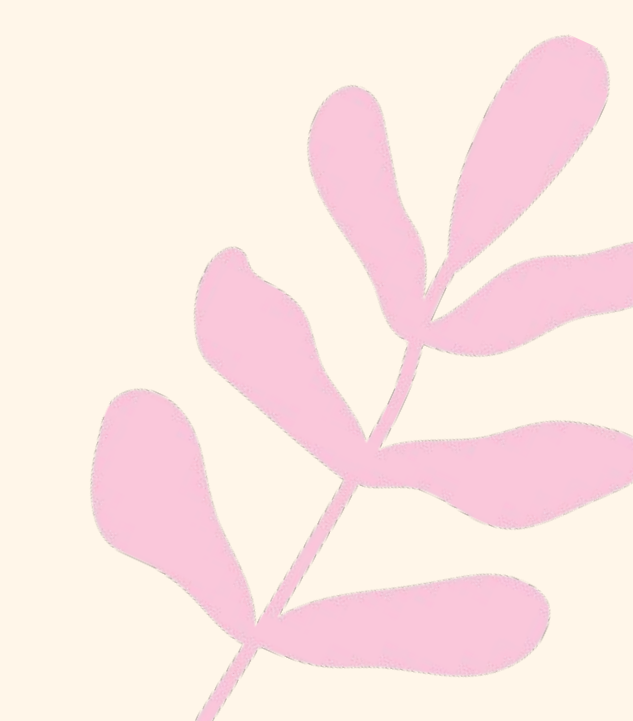
Treasure Smith

Hari Keerthi Uppalapati

Richa Kulkarni

HCIP-6108 Intermediate Decision Analysis

Dr. Tim Lowe



What is Early-Stage Triple-Negative Breast Cancer (TNBC)?

TNBC is an aggressive subtype of invasive breast cancer. In comparison to other invasive breast cancer subtypes, TNBC is characterized by faster disease progression and metastasis, fewer available treatment options, and a generally poorer prognosis.

Symptoms: new breast or underarm lump, changes in breast size or shape, skin dimpling, or nipple discharge unrelated to pregnancy or breastfeeding.

TNBC Stage Distribution:

More likely diagnosed at Stage II-III vs. HR+ cancers

Higher proportion of high-grade tumors at all stages

Stage IV: Preferentially metastasizes to visceral organs (lung, brain, liver) vs. bone

Background & Rationale

Epidemiology

TNBC comprises around 10–15% of total breast cancer cases.

The prevalence of TNBC is greater among women younger than 40 years

Age-adjusted incidence rate : 13.9 per 100,000 women in the United States (2018-2022 SEER data)

Clinical Challenge

TNBC lacks Estrogen, Progesterone, and HER2 receptors, limiting targeted treatment options

Chemotherapy has historically been the standard of care, with high recurrence risk even in early-stage disease

Early-stage TNBC still carries a substantial risk of relapse and disease progression

New Therapeutic Approach

Pembrolizumab, an immune checkpoint inhibitor, has shown improved pathologic complete response (pCR) and event-free survival (EFS) when added to standard neoadjuvant chemotherapy

Economic concern: immunotherapy substantially increases treatment costs

Rationale for CEA:

To assess whether improved clinical outcomes justify higher costs

To inform U.S. payer coverage decisions and value-based care policies

Research Question

Is neoadjuvant pembrolizumab plus chemotherapy cost-effective compared with neoadjuvant chemotherapy alone for patients with early-stage triple-negative breast cancer in the United States?

Hypothesis

- H_0 (Null Hypothesis): Pembrolizumab plus chemotherapy does not have a lower ICER than chemotherapy alone.
- H_1 (Alternative Hypothesis): Pembrolizumab plus chemotherapy has a lower ICER than chemotherapy alone.

Methods: Study Design & Perspective

Target Population

- Female adults with early-stage, high-risk triple-negative breast cancer (TNBC)
- Based on KEYNOTE-522 trial population
- U.S. patients

Perspective

U.S. healthcare payer perspective

Index Date: Date at which the patient receive first neoadjuvant treatment

Model Type: Decision Tree

Model Parameters

Cycle length: Yearly

Time horizon: Lifetime

Discount rate: 3% annually

Treatment Arms

Intervention: Neoadjuvant pembrolizumab + chemotherapy

Comparator: Neoadjuvant chemotherapy

Model Inputs Summary

Life Expectancy

State	Pembro+Chemo	Chemo
Event-Free	17.8 yrs	11.4 yrs
LR	2.5 yrs	2.5 yrs
DM	1.13 yrs	1.13 yrs

Utility Values

State	Utility
Event-Free	0.842
LR	0.768
DM	0.639

Terminal Node Total Costs (USD)

State	Pembro+Chemo	Chemo
Event-Free	\$333,348	\$126,886
LR	\$333,581	\$127,119
DM	\$382,742	\$172,268

Transition Probability

State	Pembro	Chemo
Event-Free	0.79	0.53
LR	0.08	0.16
DM	0.13	0.31
Sum	1.00	1.00

All inputs derived from peer-reviewed sources · Costs in 2025 USD · Discount rate 3% annually · US third-party payer perspective
Cost: Treatment phase + Disease management × LE + EOL costs (EOL added for LR and DM state)

Analytical Strategy

- Constructed a decision tree model in SpiceLogic Rational Will comparing two treatment arms
- Allocated patients to three health states: **Event-Free, Locoregional Recurrence, Distant Metastasis**
- Assigned utility values from EQ-5D-5L data to each health state
- Derived life expectancy from published US clinical data using exponential survival conversion
- Terminal node costs combined CMS ASP Q4 2025 drug pricing with literature-based disease management and end-of-life costs, all inflated to 2025 USD (Medical Care CPI factor: 1.127).
- Computed QALYs ($\text{Utility} \times \text{Time}$) and expected costs at each terminal node
- Produced ICER (incremental cost per QALY gained) evaluated against \$150,000/QALY WTP threshold
- Performed one-way deterministic sensitivity analysis on all 15 input variables to test robustness

Method: Life Expectancy Derivation

Life Expectancy:

Formula Used for Event Free State: $\text{Mean LE} = -T / \ln(S)$

Where: S = survival probability, T = time in years

State	Input	Calculation	Mean LE
EF (Pembro+Chemo)	84.5% EFS at 36 months	$-3 / \ln(0.845)$	17.8 years
EF (Chemo)	76.8% EFS at 36 months	$-3 / \ln(0.768)$	11.4 years
LR (Both)	Median OS = 2.5 years	Used directly	2.5 years
DM (Both)	Median OS = 13.6 months	Used directly	1.13 years*

Note: LR and DM use median OS directly from published clinical registry studies — Cahlon et al. 2019 (LR) and Huang et al. 2023 / Labros et al. 2024 (DM). Exponential conversion was applied only to EF state where Kaplan-Meier survival percentages were available .

Method: Terminal Node Cost Structure

Component	EF	LR	DM
Treatment — Pembro arm	\$225,708	\$225,708	\$225,708
Treatment — Chemo arm	\$19,246	\$19,246	\$19,246
Disease Mgmt — Pembro	\$10,764×10 \$107,640	\$36,036×2.5 \$90,090	\$86,164×1.13 \$97,365
Disease Mgmt — Chemo	\$10,764×10 \$107,640	\$36,036×2.5 \$90,090	\$86,164×1.13 \$97,365
Subsequent Therapy	\$0	\$0	Pembro: \$41,886 Chemo: \$37,874
EOL Cost	\$0	\$17,783	\$17,783
Total — Pembro arm	\$333,348	\$333,581	\$382,742
Total — Chemo arm	\$126,886	\$127,119	\$172,268

- Treatment cost appears in all states because it is paid upfront before outcome is known
- EF disease management capped at 10 years — minimal monitoring needed beyond that point
- EOL cost (\$17,783) applied only to LR and DM — patients in those states die from cancer, not EF

Cost-Effectiveness Analysis: Decision Tree

Standard Cost Effectiveness Analysis

Framework: Maximize QALY
(Utility × Time)

Minimize Cost

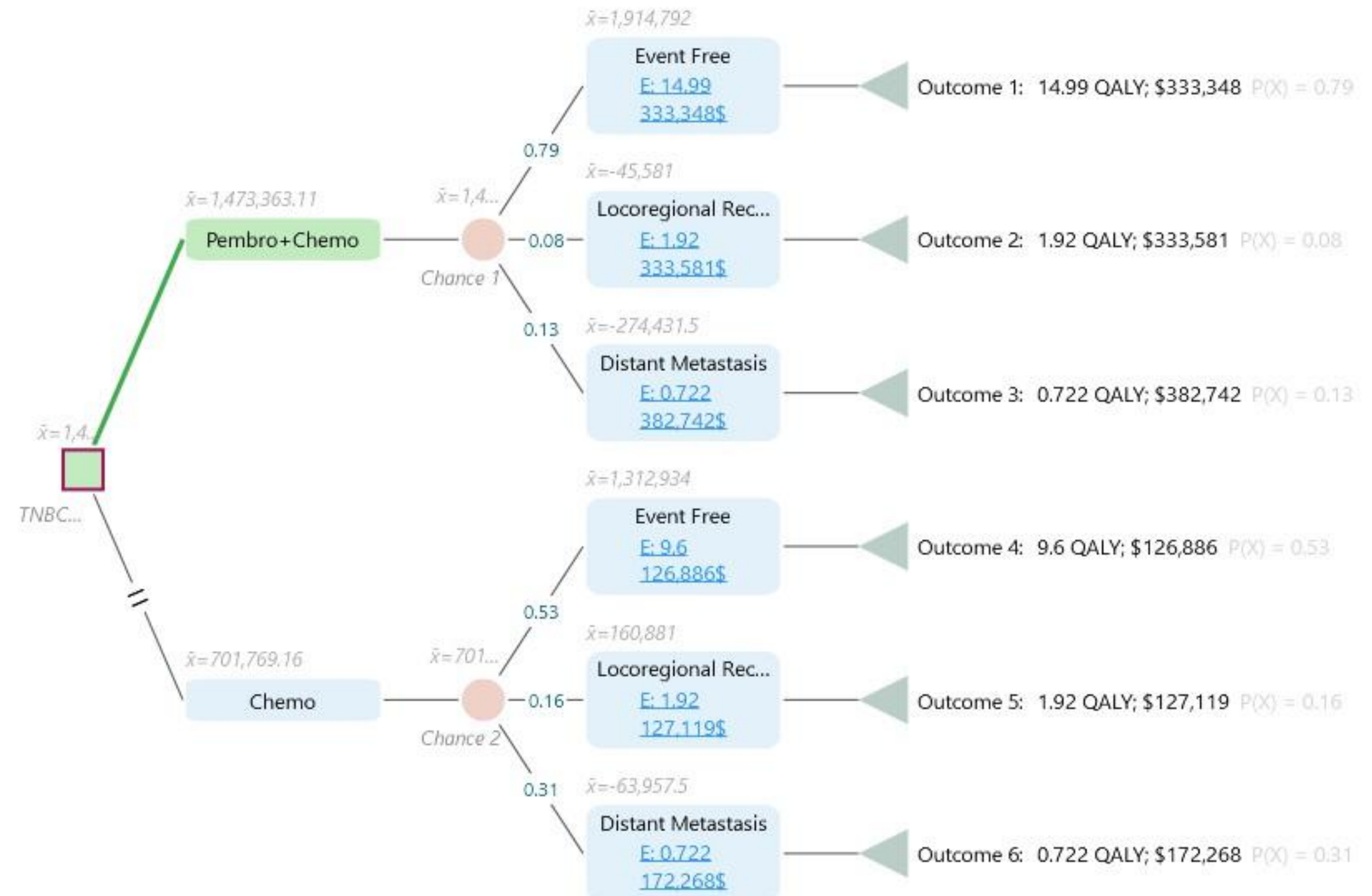
Output: ICER (\$/QALY)

Threshold: US WTP

\$150,000/QALY

Standard: US Panel on
Cost-Effectiveness
in Health & Medicine

Discount: 3% annual



Results

Rational Will Recommendation: Pembrolizumab+Chemo

Base Case Cost-Effectiveness Results:

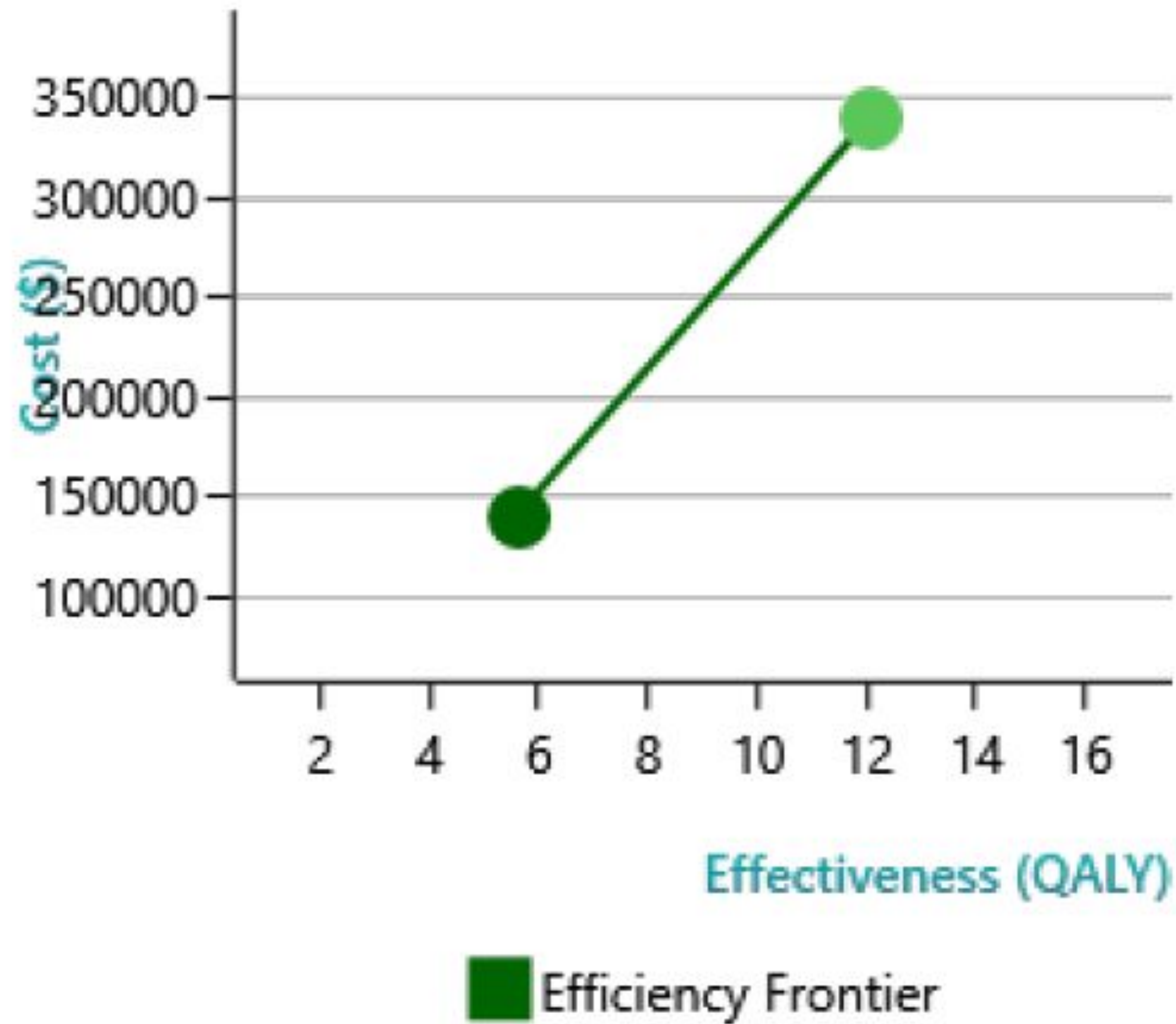
Metric	Pembro+Chemo	Chemo Only	Incremental
Expected QALYs	12.09	5.62	+6.47
Expected Cost	\$339,788	\$140,992	+\$198,796

ICER = \$198,796 ÷ 6.47 = \$30,729 per QALY gained

At ~\$30,729/QALY, pembrolizumab+chemotherapy is cost-effective — approximately one-fifth of the U.S. WTP threshold of \$150,000/QALY.

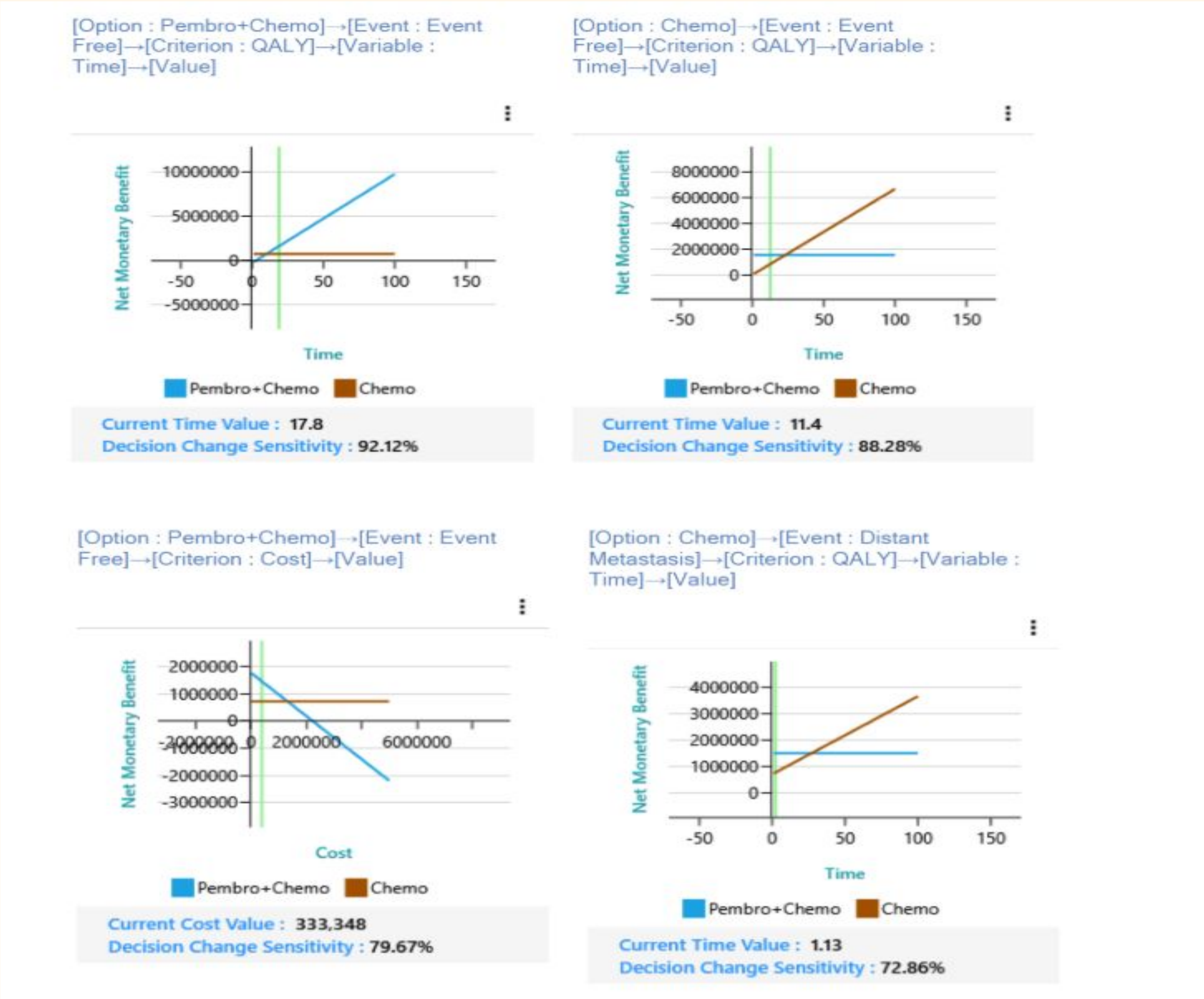
Cost Effectiveness Plane

Cost Effectiveness Plane



One-Way Deterministic Sensitivity Analysis: Result

Top 4 Most Sensitive Parameters



The four parameters that most influence the results are all linked to the Event-Free and Distant Metastasis states, showing that pembrolizumab's cost-effectiveness comes from keeping more patients disease-free and fewer patients progressing to costly late-stage disease.

Discussion

- Pembrolizumab plus Chemotherapy was cost-effective for high-risk early-stage TNBC in the US.
- ICER was \$30,729/QALY, well below the WTP threshold of \$150,000/QALY.
- Cost-effectiveness was driven by better Event-Free outcomes, with 79% remaining event-free versus 53% with chemotherapy alone.
- Despite higher upfront drug costs, the regimen generated 6.47 additional QALYs.
- Results were consistent with Huang et al. (2023), supporting robustness across model structures.
- Sensitivity analysis showed that no tested parameter change reversed the recommendation.

Strengths

- **Grounded in high-quality clinical evidence** including KEYNOTE-522 and peer-reviewed published sources such as Huang et al. (2023)
- **Transparent decision model** comparing Pembrolizumab plus Chemotherapy with Chemotherapy alone
- **Evaluates long-term outcomes** including recurrence, metastasis, survival, QALYs and costs
- **Comprehensive cost perspective** includes all major direct medical costs from drug acquisition through terminal care from a US payer perspective
- **Robust sensitivity testing** using one-way deterministic sensitivity analysis across all the input variables

Limitations

- The decision tree simplifies TNBC disease progression and does not model repeated transitions between health states over time
- Event-free life expectancy relied on an exponential survival assumption, which may not fully reflect long-term real-world survival patterns
- The healthcare payer perspective excludes indirect costs such as productivity loss and caregiver burden
- Cost estimates may change with future drug pricing, reimbursement changes, or pembrolizumab biosimilar entry
- The model does not capture time-dependent transitions or cycling between health states as a Markov model would

Conclusion

- Pembrolizumab plus Chemotherapy was the preferred and cost-effective treatment option for high-risk early-stage TNBC.
- From a US payer perspective, the regimen provided sufficient added health benefit to justify its higher cost.
- These findings support coverage and reimbursement decisions for the KEYNOTE-522 regimen.
- Future analysis should examine real-world outcomes, indirect costs, and the effect of Pembrolizumab biosimilar entry.

References

- Howlader, N., Altekruse, S. F., Li, C. I., Chen, V. W., Clarke, C. A., Ries, L. A., & Cronin, K. A. (2014). US incidence of breast cancer subtypes defined by joint hormone receptor and HER2 status. *Journal of the National Cancer Institute*, 106(5), dju055.<https://academic.oup.com/jnci/article-abstract/106/5/dju055/2607058>
- Plasilova, M. L., Hayse, B., Killelea, B. K., Horowitz, N. R., Chagpar, A. B., & Lannin, D. R. (2016). Features of triple-negative breast cancer: Analysis of 38,813 cases from the national cancer database. *Medicine*, 95(35), e4614.
https://journals.lww.com/md-journal/fulltext/2016/08300/features_of_triple_negative_breast_cancer_.38.aspx
- Lin, N. U., Vanderplas, A., Hughes, M. E., Theriault, R. L., Edge, S. B., Wong, Y. N., Blayney, D. W., Niland, J. C., Winer, E. P., & Weeks, J. C. (2012). Clinicopathologic features, patterns of recurrence, and survival among women with triple-negative breast cancer in the National Comprehensive Cancer Network. *Cancer*, 118(22), 5463–5472.<https://pubmed.ncbi.nlm.nih.gov/22544643/>
- Cleveland Clinic. (n.d.). Triple-negative breast cancer (TNBC). Retrieved February 9, 2026, from <https://my.clevelandclinic.org/health/diseases/21756-triple-negative-breast-cancer-tnbc>
- National Breast Cancer Foundation. (n.d.). Triple negative breast cancer. Retrieved February 9, 2026, from <https://www.nationalbreastcancer.org/triple-negative-breast-cancer/>
- U.S. BLS. Medical Care CPI (CPIMEDSL).fred.stlouisfed.org/series/CPIMEDSL

References

- Huang, M., Fasching, P. A., Haiderali, A., Xue, W., Yang, C., Pan, W., Zhou, Z. Y., Hu, P., Chaudhuri, M., Le Bailly De Tillegheem, C., Cappoen, N., & O'Shaughnessy, J. (2023). Cost-effectiveness of neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant single-agent pembrolizumab for high-risk early-stage triple-negative breast cancer in the United States. *Advances in Therapy*, 40(3), 1153–1170. <https://link.springer.com/article/10.1007/s12325-022-02365-1>
- Huang, M., Fasching, P., Haiderali, A., Pan, W., Gray, E., Zhou, Z.-Y., Hu, P., Chaudhuri, M., Le Bailly De Tillegheem, C., Cappoen, N., & O'Shaughnessy, J. (2022). Cost-effectiveness of pembrolizumab plus chemotherapy as first-line treatment in PD-L1-positive metastatic triple-negative breast cancer. *Immunotherapy*, 14(13), 1027–1041. <https://www.tandfonline.com/doi/full/10.2217/imt-2022-0082>
- National Cancer Institute, Surveillance, Epidemiology, and End Results Program. (n.d.). Cancer stat facts: Female breast cancer subtypes. Retrieved February 9, 2026, from <https://seer.cancer.gov/statfacts/html/breast-subtypes.html>
- Schmid, P., Cortes, J., Dent, R., Pusztai, L., McArthur, H., Kümmel, S., Bergh, J., Denkert, C., Park, Y. H., Hui, R., Harbeck, N., Fasching, P. A., Cardoso, F., Untch, M., Huober, J., Cescon, D. W., Chui, S. Y., Siguenza, M., & O'Shaughnessy, J. (2022). Event-free survival with pembrolizumab in early triple-negative breast cancer. *Journal of Clinical Oncology*, 40(7), 754–765. <https://pubmed.ncbi.nlm.nih.gov/35139274/>
- Schmid, P., Cortes, J., Pusztai, L., McArthur, H., Kümmel, S., Bergh, J., Denkert, C., Park, Y. H., Hui, R., Harbeck, N., Takahashi, M., Foukakis, T., Fasching, P. A., Cardoso, F., Untch, M., Jia, L., Karantza, V., Zhao, J., Aktan, G., ... O'Shaughnessy, J. (2020). Pembrolizumab for early triple-negative breast cancer. *New England Journal of Medicine*, 382(9), 810–821. <https://www.nejm.org/doi/10.1056/NEJMoa1910549>

Questions?

Thank You!!